

ELEMENTARY LINEAR ALGEBRA
FALL 2025 EXAM 2 OF 3

“Mathematics is not about numbers, equations, or algorithms; it’s about understanding..”
William Paul Thurston

You have worked hard this semester! Apply your mind to all you have learnt these last 8 weeks!
Give it your best.!!!

1) This is a 60 minute Exam. You may use the entire class period if you desire.

- a) Calculator allowed
- b) No notes allowed
- c) No smart devices/internet allowed

By signing below, I _____ acknowledge that I will uphold academic integrity and only use the resources allowed for each section.

Signature:_____

Date:_____

Name:_____

You may do your scratch work here.

1. a) (**10pts**) Find the basis of the null space of the matrix below.

$$A = \begin{bmatrix} 1 & -3 & 5 & 0 \\ 1 & -2 & -1 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 3 & -18 & 9 \end{bmatrix}$$

- b) (**10pts**) Find the basis of the column/range space, $(\text{Col}A)$ of the matrix A from part (a)

- c) (**8pts**) From (a) and (b), what is the dimension of the null space of and dimension of the column space of A .

- d) (**4pts**) From (a) and (b), are the columns of A linearly independent? Provide simple justification.

- e) (**3pts**) Does A have an inverse? Provide simple justification.

You may do your scratch work here.

2. a) (**5pts**) Is the set of vectors, $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}, \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 6 \end{bmatrix} \right\}$ a basis?

- b) (**10pts**) Let a basis $\mathcal{D} = \left\{ \begin{bmatrix} -6 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -3 \\ 5 \end{bmatrix} \right\}$, and $H = \text{Span}(\mathcal{D})$ is a subspace. Is $\mathbf{v} = \begin{bmatrix} -2 \\ 1 \\ 9 \end{bmatrix}$ is in the subspace, H .

- c) (**5pts**) From part (b), if $\mathbf{v}$ is in the subspace H , find the coordinate vector $[\mathbf{v}]_{\mathcal{D}}$.

- d) (**10pts**) Let $\mathcal{B} = \{(0, 5), (3, 1)\}$ and $\mathcal{C} = \{(1, 2), (-1, 3)\}$. Find the matrix that transforms vectors from the $\mathcal{B}$ -coordinates to $\mathcal{C}$ -coordinates, $P_{\mathcal{B} \rightarrow \mathcal{C}}$.

- e) (**Optional** $\rightarrow$ **Bonus: 5pts**) From (d), if the transformation matrix of T in the $\mathcal{B}$ basis is:

$$A = \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix},$$

what is its representation within the $\mathcal{C}$ basis coordinates? **Read the question carefully before attempting**

3. a) (**25pts**) Determine which of the following vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$, and $\mathbf{v}_5$ are eigenvectors of the matrix B .

$$B = \begin{bmatrix} 6 & -1 & 0 \\ -1 & 6 & -1 \\ 0 & -1 & 6 \end{bmatrix}$$

If $\mathbf{v}_i$ is an eigenvector, simply state "**YES**", and give the corresponding eigenvalue. Otherwise, state "**NO**". Show your work!

i) $\mathbf{v}_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$

ii) $\mathbf{v}_2 = \begin{bmatrix} 1 \\ \sqrt{2} \\ 1 \end{bmatrix}$

iii) $\mathbf{v}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

iv) $\mathbf{v}_4 = \begin{bmatrix} 1 \\ -\sqrt{2} \\ 1 \end{bmatrix}$

v) $\mathbf{v}_5 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$

- b) (**10pts**) Compute the matrices P and D such that the matrix B from part (a) is diagonalizable (that is, $B = PDP^{-1}$).

Great job for all your hard work!